

Zhang Heng

One of the greatest scientists in Chinese history, Zhang Heng combined being an inventor with a career as a government official. He rose to become China's chief astronomer, as well as an important advisor to the Emperor. As if this was not enough, Zhang was also a famed mapmaker, poet, and artist.

WATER-POWERED ARMILLARY SPHERE

An armillary sphere is a framework of rings that was used to understand how planets and stars move through the sky. Zhang's big idea was to use a waterwheel to power the sphere's machinery so that it turned automatically, making one rotation a year.

Replica of Zhang Heng's armillary sphere, 1439



Rings were positioned to mirror the movement of objects in space.

▼ EARTHQUAKE INDICATOR

The most famous of Zhang's inventions was his seismoscope, or "earthquake weather vane," that could tell the direction in which an earthquake had occurred.



Each dragon head faces a different direction.

Bronze urn

Pendulum inside drops a ball through the dragon's head facing the direction of the quake.

NAVIGATIONAL CHARIOT

Zhang may have also invented a device to indicate direction. This was a "chariot" with a figure on top that could be positioned to point in any direction. A complicated system of gears meant that no matter what path the chariot followed it continued to point in the same direction, similar to the way a compass does.

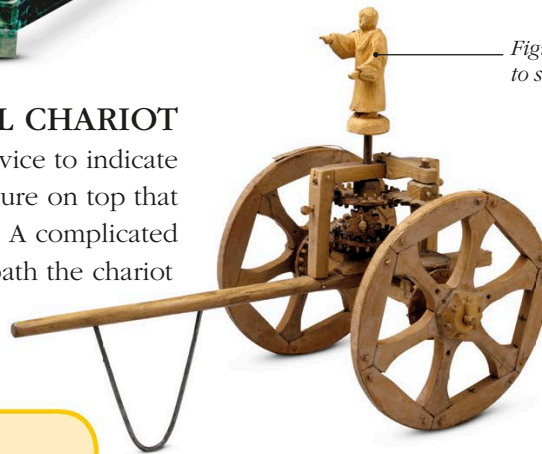


Figure always points to set direction.

Model of a navigational chariot from China, 2700–1100 BCE

LIFE STORY

78 CE

Zhang Heng is born near Nanyang, a city in central China. His father died when Zhang was just 10 years old.

95 CE

He leaves home to go to Luoyang, China's capital at the time, to study at the Imperial Academy.

108 CE

While working as a local official, he begins to publish papers about astronomy and mathematics.

112 CE

Zhang Heng is summoned by the Emperor to work as an official at the imperial court in Luoyang.

